

Proof of genuine engagement: blockchain and the neo-slacktivism problem

This article examines AI-driven neo-slacktivism, i.e. superficial digital engagement increasingly generated or amplified by large language models and algorithmic content recommendation systems, and interrogates how blockchain distinguishes genuine human civic participation from synthetic manipulation in virtual spaces. Relying on field notes with blockchain developers, social media practitioners and data governance experts, and situated within institutional theory, the study proposes cross-referenced civic ledgers, cryptographic tokens and metadata identifiers as mechanisms tying digital interactions of humans to verified identities, generating trust at lower cost and rendering sustained automated engagement, i.e. token exhaustion, economically prohibitive. Findings indicate that blockchain offers a promising accountability layer, yet its effectiveness remains contingent with transparent governance, regulatory clarity and public awareness. As an agenda setting piece, the article situates blockchain not as a substitute for human trust but as its verifiable complement that depends on organizational and institutional frameworks it is embedded.

Keywords: blockchain; AI-driven neo-slacktivism; decentralized social networking; civic engagement; metadata identifiers

1. Introduction

1.1 Conceptual definitions: from conventional slacktivism to AI-driven neo-slacktivism

Slacktivism, also called clicktivism (Karpf, 2010) or armchair activism (Cabrera, Matias & Montoya, 2017), refers to minimal effort digital engagement with social or political causes (e.g. via social media, e-democracy or e-petition platforms) that carries no real world commitment and exists purely in virtual space. Coined in 1995 by Dwight Ozard and Fred Clark to describe small scale, positive personal actions, the term later acquired a pejorative connotation (Morozov, 2012), denoting low effort, performative gestures such as hashtag sharing, liking, light commenting or signing digital petitions. In this regard, slacktivism, as a synthesis of two opposing cultural concepts, i.e. slackerdom and activism, reflects both the convivial affordances of modern digital communication and the distinctive, ambivalent behavioral habits which humans exhibit within social networking settings.

Early scholarship on the phenomenon was largely dismissive, treating such low threshold behaviors as narcissistic, feel-good gestures that diluted political efficacy by promoting the illusion of activism without its associated risks or costs (Shirky, 2008). However, subsequent longitudinal and network-based studies have caused a significant conceptual shift, reframing slacktivist engagement not as a substitute for offline activism but as a vital tool that lowers entry barriers for historically marginalized populations and drives agenda-setting and consensus mobilization (Skoric, 2012; Christensen, 2012; Zohouri, Darvishi & Sarfi, 2020). These low effort inputs are now understood as somewhat welcoming onboarding mechanisms within connective action models (Leonel et al, 2024), building the solidarity and communication infrastructure necessary for large scale offline mobilization, particularly among youth and marginalized groups and also in repressive contexts where such engagement may constitute the

only feasible form of dissent (Lee & Hsieh, 2013; Wike et al, 2022). As a result, many activists now regard the term slacktivism itself as dismissive, though caution still remains regarding performative engagement, yielding a more pedagogically nuanced understanding of its potential, limitations and complexity within broader political processes (Glenn, 2015; Ralston, 2024). In addition, the debate now further complicated by the growing role of low effort AI-driven interactions in the social media environment.

Politically, slacktivism denotes low threshold digital engagement through which support for public causes is signaled, transforming social media users, semantically, from passive participants into co-creators (Baskerville & Myers, 2023), indeed an important element of modern political socialization. However, the recent advent of large language models (LLMs) means that such humanlike content can now be readily generated and proliferated by artificial intelligence systems, including through automated content recommendation algorithms (Kassen, 2026a). According to Amazon researchers, 57% of internet communication content is already likely AI-enhanced, with the trend accelerating rapidly (Thompson et al, 2024), a pattern arguably lending credence to the recently proposed Dead Internet Theory (Walter, 2024), which holds that online interaction will progressively become predominantly artificial, rendering genuine human communication increasingly anomalous.

Alongside cyberslacking (Osei, Ampofo & Osei, 2022), i.e. the use of work provided internet access by employees for personal, non work related activity, slacktivism is also associated with the emerging phenomenon of digital conviviality (Yékú, 2022): a sociobehavioral movement, industry supported, oriented toward shared purpose and mutual benefit rather than the hierarchical, profit-driven logic of social networking platforms (thus paradoxically all offered as a service free of charge by the corporate sector). In practice, however, its realization amounts merely to a comfortable, AI-optimized user environment engineered for smooth interaction, giving rise to a new algorithmic slacktivism, or neo-slacktivism (Kassen, 2025; 2026b), i.e. superficial, low effort online engagement amplified and manipulated by machine learning algorithms.

Widespread digital communication indeed eases interactions between people and raises public awareness of political, economic and social processes, including in organizational contexts (Kuusisto, 2017). Yet the low cost of such networking also drives platforms toward armchair activism as a communicative strategy, a pattern further exacerbated by the proliferation of machine learning-driven content recommendation systems, disinformation and other tools of public mind manipulation that decrease trust across online communication channels (Werenowska & Jaska, 2025). Addressing neo-slacktivism, therefore, requires first re-conceptualizing slacktivism as an intrinsically multidimensional phenomenon, one whose manifestations have often been dismissed as exaggerated or peripheral relative to its predominantly positive implications for social media-based engagement (Christensen, 2011; Dennis, 2019), including its potential for more effective advocacy and e-participation (Zohouri, Darvishi & Sarfi, 2020), that is a shift demanding that academic perspectives move decisively beyond their initial skepticism.

1.2 AI-driven neo-slacktivism as a phenomenon: why is it a concern?

The artificial intelligence industry currently faces a bottleneck known among data governance practitioners as the Data Wall, whereby large language models have essentially exhausted the accessible supply of high quality, human-generated text. To sustain continued expansion, developers increasingly resort to synthetic data (WEF, 2025), i.e. content generated by existing models to train their successors. However, heavy reliance on such uncharted synthetic outputs, often termed slop, introduces a well-recognized mathematical risk: training model collapse, or AI cannibalism and inbreeding. When a system is repeatedly trained on prior machine outputs, errors, statistical biases and hallucinations compound exponentially across generations, yet rare facts, nuanced stylistic expression and niche human knowledge are gradually averaged out. The process, not unlike photocopying a photocopy, basically threatens to transform promising versatile algorithms into somewhat sterile, repetitive systems trapped within self-referential synthetic data ecosystems.

To counter this diminishing supply of authentic human data, arguably the ultimate oil of contemporary technological innovation, yet increasingly synthetically manipulated (Tarsney, 2025) and paradoxically corrosive of human autonomy (Catena, 2026), social media platforms must proactively incentivize genuine user participation, even where such engagement initially takes the lightweight form of clicktivism. In this regard, individual users are now targeted through AI-driven neo-slacktivist tools such as metadata-informed content recommendation algorithms, which infer behavioral habits from predictive analytics in an intrinsically centralized manner. Metadata, i.e. the hidden layer of contextual information describing user profiles, posts, networking interactions, identity and behavior, functions as an invisible infrastructure that people rarely notice yet that underpins the modern digital social networking world, currently harvested through various centralized, machine learning-driven ecosystems (see table 1).

Table 1. Types of metadata automatically collected by traditional centralized social media platforms

#	Category	Examples of metadata	Purpose
1.	User profile	Username, bio, location, join date, profile links, verification status, email, phone number, address, country, city	Identifies and connects users to others and content they create, share or interact with
2.	Content	Timestamps, geotags, hashtags, language, post identification, devices used	Organizes posts and tailors search engines and trends they monitor
3.	Engagement	Likes, shares, comments, saves, view counts, click-through rates, content ranking, reports, etc.	Measures reach and fuels algorithmic ranking
4.	Behavioral	Time spent on posts, accounts followed, ads clicked, block, mute or hide actions in regard to peers and their content	Profiles user interests for content targeting
5.	Network	Friends, followers, groups joined, tagged relationships, blocked contacts	Maps social connections and networking analysis for more tailored recommendations
6.	Technical	IP address, browser type, operating systems, cookie data, app permissions	Tracks activity across devices and platforms

7.	Biometric	Unique usage patterns for user face and fingerprint identification, patterns of voice control use, in app lock, pulse or heart beat patterns	Identifies user behavior, concerns, including medical ones and thus vital needs across devices or patterns of behavior
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Furthermore, these trends can be artificially inflated through automated expert information systems, including for data management practices (Sposato, 2025) and social networking fraud (AIID, 2025), necessitating a more contextual approach to theorizing the political and socioeconomic implications of slacktivated knowledge sharing (Seitz & Misra, 2020). Armchair activism thus warrants renewed scrutiny as a sociobehavioral, political and socioeconomic phenomenon, encouraged by social media commercial models built on AI-enhanced interaction, hence neo-slacktivism: a new digital realm in which public attention and reaction function as socioeconomic currency, actively monetized as the backbone of such data-driven business models in industry. In this regard, the commercialization of public attention drives such digitalization trends, since human reactions in social media constitute a key source of marketing (Cao & Belo, 2023), financial (Liang, 2022), political (Brainard, 2016), public (Seigler, 2017), corporate (Liou & Hu, 2019) and other valuable, including big, data (Hu, 2019), which predominantly centralized harvesting platforms are positioned to exploit further, particularly when advertising such content discreetly (Xu, Chen & Ow, 2023; Shao et al, 2023) alongside the emotional dimensions of neo-slacktivism, effects especially pronounced within online health communities (Zhou et al, 2023). AI-enhanced neo-slacktivism, given its low cost of application, thus facilitates these trends, simultaneously generating distorted public feedback from civic, governmental and corporate stakeholders, further undermining the rather utopian assumptions once made about effects of social media on the efficiency of public organizations (Brainard, 2016).

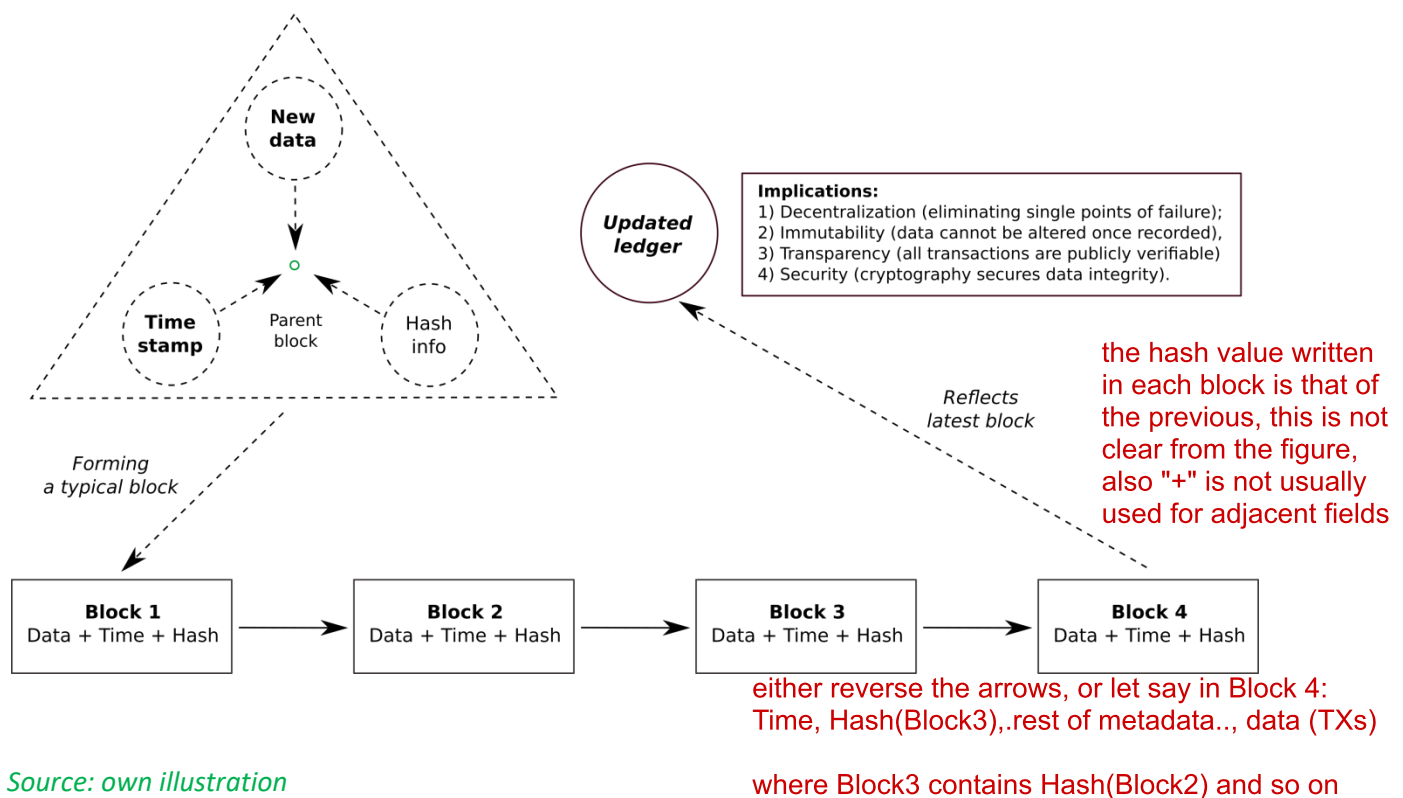
One wedge, however, may drive out another: AI, and blockchain-enhanced AI in particular, holds potential to address the malevolent manifestations of neo-slacktivism. Disinformation and data manipulation can be specifically countered in social media settings through the capacity of blockchain to generate public trust, **both conceptually and practically (Zhan, Xiong & Xing, 2023; Kassen, 2026b; Huang & Zhan, 2026)**, with distributed ledger technologies promising more sophisticated, incentivized and reliable means to endorse, promote and verify public interaction within a decentralized social networking environment (Hisseine, Chen & Yang, 2022; Buțincu & Alexandrescu, 2023; Kim, Huh & Kim, 2024). Though largely undertheorized academically, this approach could fundamentally redefine and verify digital engagement, transforming slacktivism and its transient gesticulations into practically feasible pathways for tangible impact.

2. Blockchain and its affordances: understanding implications for neo-slacktivist strategies

Conceptually, blockchain is widely regarded as a disruptive participatory technology (Frizzo-Barker et al, 2020; **Wang, Schlagwein & Seymour, 2025**): a decentralized digital ledger, part of the broader distributed ledger technology family, that records transactions in blocks linked chronologically and cryptographically, and is thereby transparent, secure and immutable. Four features render it unique within the modern digital landscape: decentralization, immutability, transparency and security. Unlike traditional databases reliant on a central authority, blockchain is distributed across a network of peers,

meaning no single entity, public or private, controls the ledger, rendering it more robust to tampering and censorship; once a block is added to the chain, it cannot be altered or deleted without network-wide consensus. Transparency, in turn, generates trust without human proxy, since every transaction on a permissionless public blockchain is publicly viewable, building public confidence in decentralized information systems generally (Sari, Ilmananda & Romano, 2021) and in tokenized social media specifically (Li, Loahavilai & Naktnasukanjn, 2022; Alizadeh, Hoes & Gilardi, 2023) by enabling scrutiny of recorded transaction data (Guidi, 2020; Salau et al, 2022). Cryptography, finally, secures the integrity of the ledger by linking each block to its predecessor through a cryptographic hash (CH) and timestamp (TS): the network of peers verifies transaction data (TD) before appending it, and since each block depends on the one before it, altering any single block would require recomputing every subsequent block in strict chronological sequence, a task rendered computationally prohibitive (Li et al, 2021) absent majority network consensus, itself governed by the consensus algorithm and proof-of-computation model chosen, which may vary considerably depending on the administrative goals or governing philosophy of the network's owners, particularly where the ledger is permissioned (see figure 1).

Figure 1. Understanding the conceptual mechanism of blockchain-driven data management



Source: own illustration

The two most widely regarded standard consensus mechanisms (Sriman, Ganesh Kumar & Shamili, 2021) are proof of work (PoW) and proof of stake (PoS): fundamentally different algorithms used to validate transactions and secure the network cryptographically. PoW requires miners to solve computationally intensive mathematical problems, whereas PoS selects validators according to the crypto assets they stake as collateral, offering a more energy efficient mechanism in which the more

tokens a user stakes (e.g. politically via e-participation tokens, or economically via cryptocurrencies), the greater their chance of validating the next block and earning further rewards.

A further, increasingly promising development to ensure genuine participation in decentralized social media is the emergence of Proof of Humanity (PoH) systems. Unlike PoW or PoS, which govern how networks validate transactions, PoH operates as an application-layer identity verification protocol built atop existing infrastructures such as Ethereum, aiming to establish that each blockchain account corresponds to a unique human being, yet preserving pseudonymity and avoiding centralized identity providers; the Democracy.Earth project (DEEP, 2025) offers one of the best-known implementations, wherein participants submit evidence (e.g. a short video) that is reviewed through decentralized community verification, yielding a publicly verifiable, on-chain credential. PoH thus functions as a decentralized registry of verified human identities, usable by decentralized applications, groups, data governance platforms and, potentially, blockchain-driven social networks.

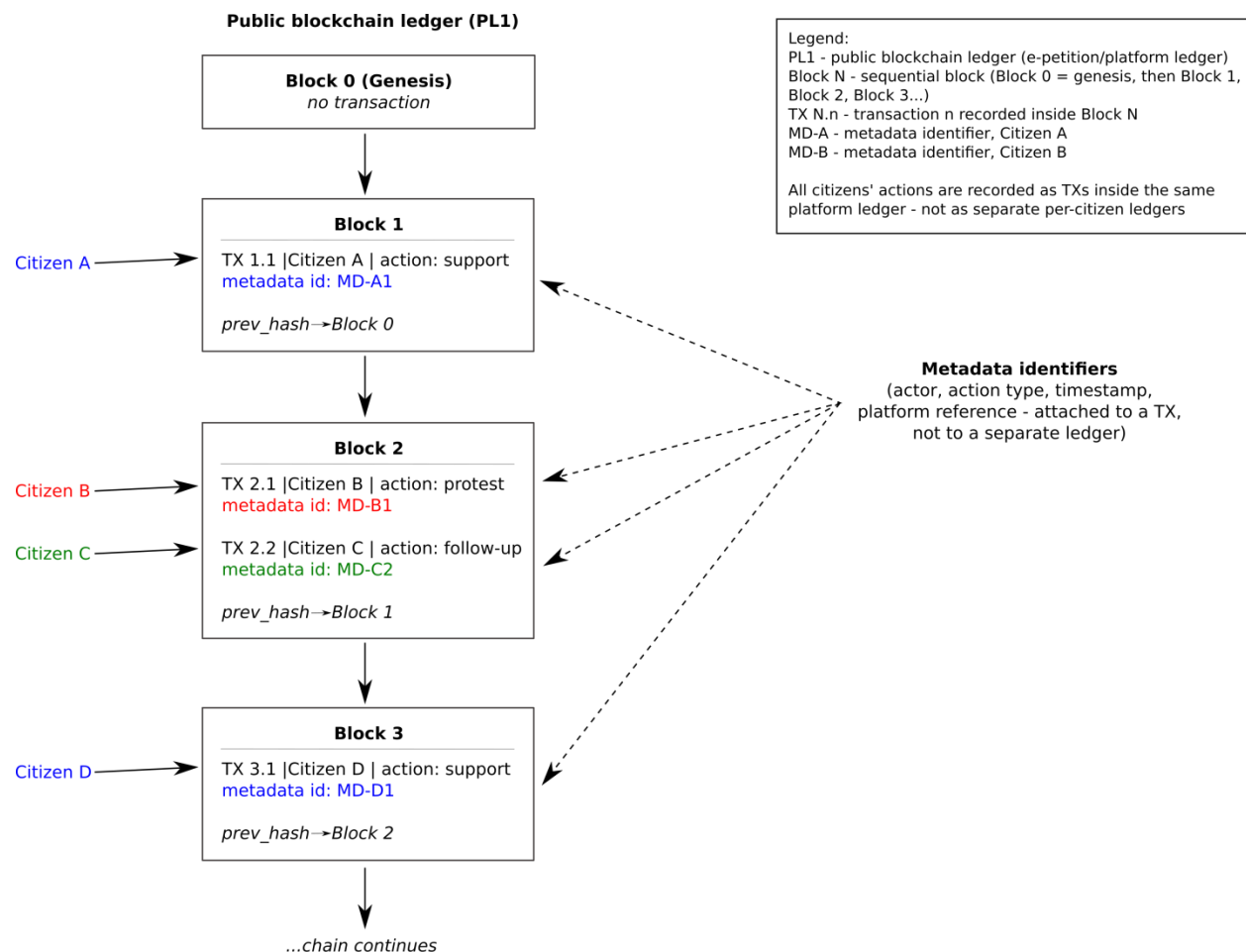
In practice, blockchain-driven information management is typically organized around a decentralized autonomous organization (DAO): an organization managed by its members through a decentralized, trustless process, decision making facilitated by distributed e-voting mechanisms and governance tokens (Wang et al, 2019). Often described as a crypto network (Song, Pei & Li, 2023) or cooperative (Alamsyah, Kusuma & Ramadhani, 2024), the DAO operates without central authority or hierarchy, relying instead on smart contracts, i.e. self-executing code that automatically enforces rules and triggers actions once predefined conditions, dependent on the chosen consensus mechanism, are met.

Beyond its association with cryptocurrencies such as Bitcoin and Ethereum, blockchain offers numerous other applications benefiting from automated code execution across public and corporate sectors (Ressi et al, 2024; Viano, 2024), including: supply chain tracking (e.g. tracing products from origin to consumer, monitoring environmental footprints or pesticide use); digital identity management (e.g. secure e-passports and cross-border document certification, decentralized e-customs); e-voting systems (e.g. secure, transparent digital voting for political elections, e-petitions and other e-participation platforms); public record keeping (e.g. secure storage of medical, military or other classified records); and smart contracts more broadly (e.g. automating agreements between stakeholders, e-notaries, secure transactions).

By leveraging its core strengths, i.e. transparency, immutability and tokenization, public activist movements could construct effective metadata-driven ecosystems atop blockchain (Freni, Ferro & Ceci, 2020; Salau et al, 2022; Li, Loahavilai & Naktnasukanjn, 2022; Alizadeh, Hoes & Gilardi, 2023). Previous research cautions against assuming benefits or risks of blockchain are inherent to the technology itself, urging instead its examination as a socio-technical system requiring careful governance attuned to emerging trends and impacts (Shin & Ibahrine, 2020). Within such a system, low effort online support, e.g. a like, repost or comment, functions merely as an initial, immutably recorded entry point, rendering the whole process quantifiable and cryptographically verifiable through unique identifiers (Kassen, 2025; 2026a), potentially helping to authenticate legitimate transactions and flag synthetic interactions, thereby addressing the malevolent effects of AI-driven neo-slacktivism. Schematically, one could

imagine a platform where signing an e-petition or sharing campaign content generates a verifiable, on-chain credential, positioning the user at the center of the system (Guidi, 2020; Guidi & Michienzi, 2020; Ba, Zignani & Gaito, 2022); this credential, or metadata identifier, could then serve as a prerequisite or multiplier for subsequent, much higher impact actions tracked on the same civic ledger, e.g. donating to a transparent blockchain-managed fund, earning tokenized rewards for verified offline tasks (such as attending a geolocation-validated local event), or gaining governance rights within decentralized activist organizations based on demonstrated contribution tiers. Such cross-referenced identifiers could thereby help DAOs address AI-driven neo-slacktivism specifically, by authorizing and recording only human-verified interactions via unique civic tokens (see figure 2).

Figure 2. The conceptual mechanism of blockchain-driven cross-referenced ledgers and metadata identifiers



Note: block numbering follows standard blockchain convention (Block 0 = genesis, then sequential integers). Sub-numbering (e.g. "1.1") is used only for TX identifiers within a block.

Source: own illustration

These metadata identifiers create an immutable contribution graph that moves beyond the vanity-metrics ecosystems of traditional social media, establishing a new denominator, a traceable e-participation token (Benítez-Martínez, Hurtado-Torres & Romero-Frías, 2021), of verifiable and escalating engagement. Smart contracts could accordingly unlock resources or tokenized recognition only once predefined thresholds of verified transactions, beyond simple sharing or reposting, are collectively met. Although accessibility and environmental concerns remain valid, blockchain offers a technological framework capable to combat inherently performative neo-slacktivism by making inaction visible, rewarding depth of interaction over breadth, and creating a ledger that distinguishes genuine contribution from hashtag activism and other neo-slacktivist gestures, namely a potentially transformative avenue that merits deeper investigation given the unprecedented recent growth of AI-driven data management ecosystems.

This research accordingly aims to highlight unique potentials of blockchain, i.e. elevated accountability and disruptive traceability mechanisms, to ensure the integrity of recorded social networking data, whereby even reversed or withdrawn actions (e.g. a subsequently deleted like or comment) remain part of a verifiable historical record, enabling more accurate auditing of engagement patterns, discouraging manipulation through coordinated retraction cycles, and providing a basis for accountability unavailable to purely mutable systems. Equally significant is its capacity to enable anonymity or pseudonymity, verifying a single, unique identity per participant without revealing that identity, thereby encouraging fearless civic participation free of retaliation or surveillance, yet still addressing bots and coordinated inauthentic behavior of users who may use them. This positions this article as agenda setting, focused on technological and organizational denominators to measure and incentivize genuine human impact, and on understanding the unique capacity of blockchain to enhance authentic activism and civic engagement in online settings increasingly shaped by AI-driven tools, distributed ledgers, metadata identifiers and other blockchain-enabled e-participation solutions.

3. Understanding methodology of the research

3.1 Organization of interview meetings

The research relied on semi-structured interviews to gather nuanced insights into the intersection of blockchain, social media and data governance. In this regard, respondents for the study were selected mostly through purposive sampling, limited to individuals with relevant expertise in blockchain development, social networking technologies and data governance, identified via targeted keyword searches on GitHub (“blockchain,” “social networks,” “distributed ledger,” “data governance,” “information management”), which yielded an initial pool of 56 experts. To reduce potential bias, individuals directly involved in cryptocurrency trading and mining were excluded, since their interests might have skewed findings toward financial rather than social and governance implications of blockchain as a technology. Of the 56 candidates identified, 22 agreed to participate: 17 completed interviews online and 5 offline during a first wave conducted between July and August 2024. A second cohort of nine additional participants, primarily representing governance and social media expertise, was recruited in April 2025, bringing the total interview sample to 31 participants across the two waves.

Interviews, conducted primarily online via Zoom to allow international participation and scheduling flexibility, followed predefined open-ended themes, capturing both technical insights (e.g. consensus mechanisms, metadata identifiers and ledger architectures among others) and broader sociopolitical reflections on trust, regulation and digital participation.

3.2 The organization of focus group discussions

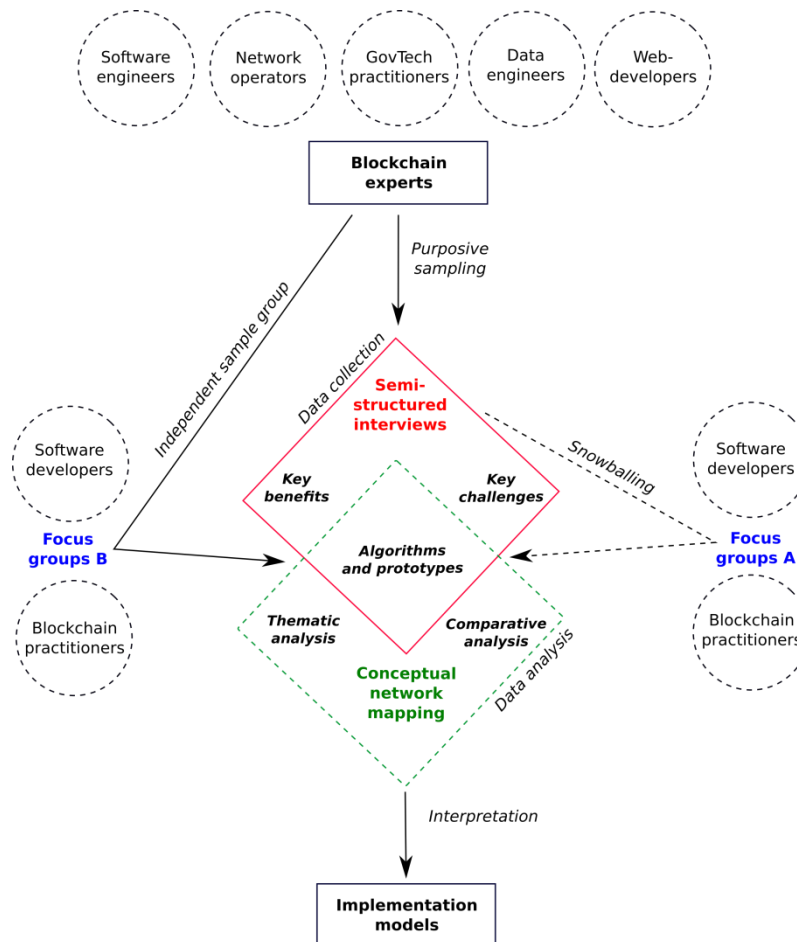
To complement the individual interviews, the research conducted two rounds of focus group discussions exclusively with software developers and practitioners. The first, convened in December 2024, comprised 14 participants who had previously completed interviews; the second, held in June 2025, included 12 newly recruited participants who had not taken part in the individual interviews, introducing additional technical perspectives to further examine models and algorithms emerging from earlier stages of the research. The two focus groups thus involved 26 participant places in total. Each session was conducted virtually via Zoom, with breakout rooms enabling smaller discussions for specific algorithmic models, consensus protocols and use cases for decentralized social networking; the first and second sessions lasted 129 and 110 minutes, respectively. In this regard, WhatsApp groups established before and after each session facilitated continuing dialogue and clarification of issues raised. The groups were structured to encourage collaborative critique through scenario testing of blockchain adoption models, peer-to-peer metadata identifiers and consensus mechanisms, supporting peer feedback and examination of practical barriers to adoption, including platform outreach, regulatory compliance, usability, public distrust of blockchain, reputational risks and potential misuse of blockchain-based systems in activist contexts. By incorporating IT experts who had not previously participated in the interviews, this stage corroborated interview findings concerning the potential of blockchain to address algorithmic neo-slacktivism and its limitations.

3.3 Analysis of research data and sampling approaches

The study employed thematic analysis, stakeholder comparison and conceptual network mapping to interpret the collected data. Thematic analysis identified repetitive patterns within interview and focus group transcripts, organized around five research questions: the capacity of blockchain to track genuine human activity, data management techniques, innovations for building trust, models of technology adoption and systemic risks, providing a structured link between expert narratives and broader conceptual models of technology adoption, including differentiated consensus mechanisms. Stakeholder analysis further enriched interpretation by comparing perspectives across developers, social media specialists and governance experts; developers, for instance, emphasized platform outreach and open-source frameworks, whereas governance practitioners stressed regulatory adaptation and ethical concerns. Conceptual network mapping then organized relationships among technical and social elements identified through the qualitative analysis, including peers, civic ledgers, metadata identifiers, cross-referenced records and forms of decentralized engagement, yielding maps and illustrative diagrams that were interpretive and conceptual rather than statistical. In this regard, these analytical approaches in combination provided a multidimensional perspective connecting technical feasibility of

blockchain with its broader sociopolitical implications to address neo-slacktivism via the algorithmic models and prototypes identified, discussed and refined (see figure 3).

Figure 3. Methodological framework for data collection, analysis, and interpretation



Source: own illustration

3.4 Coding of data: understanding key approaches and processes

The coding of research data followed an abductive approach to ensure both theoretical basis and openness to emergent insights. In this regard, deductive codes derived from the conceptual framework of the study, particularly affordances of blockchain (e.g. transparency, immutability, tokenization and decentralization among others), whereas inductive codes emerged from unexpected expert narratives, e.g. apparent skepticism about public readiness or concerns over technological challenges, particularly the assumption that blockchain alone can resolve complex problems of trust, misinformation and digital manipulation. As such, data were coded to capture both technical mechanisms (e.g. proof of work versus proof of stake protocols, metadata identifiers, smart contracts, etc.) and social dimensions,

including trust generation, digital exclusion and the potential commercialization of public attention; coding was also stakeholder relative, distinguishing pragmatic concerns of blockchain developers (e.g. platform outreach or latency issues) from emphasis of governance experts on accountability gaps and regulatory issues. In addition, negative case scenario analysis helped to address paradoxes such as the capacity of blockchain to both combat and exacerbate neo-slacktivism, depending on adoption context with visual conceptual layouts structured around themes and illustrative frameworks, clarifying in detail the important role of metadata identifiers and civic ledgers in the process.

4. Theoretical foundation of the research

Institutional theory provides a foundation to understand blockchain adoption amid AI-driven neo-slacktivism, highlighting how organizations respond to coercive, normative and mimetic pressures to gain legitimacy (Melitski, Gavin & Gavin, 2010; Rastogi & Pandita, 2026). Social media platforms typically rely on centralized, profit-driven logics that prioritize engagement metrics, inadvertently incentivizing superficial interactions, whereas blockchain challenges these logics via new legitimacy criteria, i.e. traceability, transparency and resistance to manipulation, introducing what could be framed as trustless legitimacy, where trust is generated by cryptographic accountability and not only centralized authority. Institutional inertia nonetheless complicates adoption: coercive pressures, such as fragmented regulations (Alamsyah et al, 2026), lag behind technological advances; normative pressures, shaped by user expectations of convenience, remain incompatible with complex decentralized models and would require new legal frameworks (Grech et al, 2025); and mimetic pressures, particularly the dominance of agile AI algorithms (Simon, 2024), reinforce reliance on centralized solutions. In this regard, interviewed blockchain experts accordingly emphasized institutional agility, i.e. experimental regulatory projects, open source frameworks and civic literacy programs, as necessary to normalize decentralized interactions among broader audiences. Blockchain thus stands as a poster child of this wider struggle to accommodate legal standards in digital communication: absent deliberate institutional innovation, it may remain merely an idealistic narrative for proponents of digital democracy.

5. Blockchain and tracking digital activity of humans in social networks

This section is like a long essay, should organized into several subsections, could keep the same title or change it to something like "Blockchain in Social Networks"

5.1 Blockchain Role & Limitations make it Role only, and discuss the limitations with more details in sec.10 especially if you can map to the example failures

One of the main challenges in tracking and enhancing genuine human interactions online concerns the limited availability of tools capable to record and verify such activity effectively, that is a challenge blockchain cannot resolve independently. As Shin and Ibahrine (2020) argue, blockchain must be examined as a socio-technical assemblage in which technological capabilities interact with social, cultural, institutional and governance conditions; from this perspective, blockchain can record, trace and verify the provenance of digital actions and public commitments, but whether a contribution is genuine, meaningful or socially valuable depends on the rules, identity, reputation, incentive and even penalty mechanisms and data governance arrangements surrounding the system, thus they are all developed, earned and maintained by real humans: developers, moderators and users.

*“Transparency is a key promise. Transactions are traceable, so are the people’s reactions to them [...]. Recorded information is unique, non-fungible. Records are permanent”
(Interview 9)*

This distinction matters particularly in the context of digital symbolism, where public engagement may be reduced to easily measurable interactions such as likes, dislikes or reposts, including activity generated or amplified by automated systems. Blockchain-based and decentralized social architectures, including protocols such as [\(Lens.xyz\)](#) and [\(Farcaster.xyz\)](#) as in next page, can complement these conventional metrics by maintaining more persistent and verifiable records of user activities and their associated identities or social relationships. In particular, blockchain-based components can provide auditable records of selected interactions where decentralized social graphs can preserve activity histories across applications, i.e. not solely under the control of a single platform (Guidi & Michienzi, 2020; Freni, Ferro & Ceci, 2020; Viano, 2024; Agrawal et al, 2024). Nevertheless, such mechanisms do not automatically distinguish meaningful from superficial or strategically motivated activity; their effectiveness depends on how participation is defined, incentives and penalties designed, reputation established and governance arrangements structured to address manipulation and unequal engagement. Blockchain thus enhances verifiability and traceability, but cannot by itself guarantee authenticity or social value.

“Tokens are universally used in the industry as a golden standard to properly verify such activities. I mean it is a representation of value.” (Interview 14)

5.2 Blockchain Choice (Permission vs. Permissionless)

Although many experts acknowledge that permissionless ledgers offer substantial transparency and decentralization by immutably recording interactions on a shared, open ledger, they face significant practical limitations in complex environments of networking or communication, including slow transaction speeds, high fees, privacy concerns from public ledger visibility, and compliance challenges around retrospectively modifying or removing data. These hurdles help explain why many enterprises favor permissioned architectures, including Hyperledger-based systems (Li, Wong & Guo, 2020), which sacrifice some openness and decentralization for greater control, performance and regulatory compatibility. This trade-off, however, should not be read as evidence that permissioned systems are universally preferable: the anonymity or pseudonymity of permissionless systems may prove particularly valuable to human rights activists, dissidents and other users requiring protection from institutional surveillance or retaliation. The choice between permissionless and permissioned architectures therefore depends on the purpose and social context of the application, naturally enterprise and sustainability-oriented systems favoring controlled participation and operational efficiency (e.g. Fabric or Besu), activist and human rights oriented applications favoring openness, censorship resistance and anonymity (e.g. Hyperledger Indy or Iroha) which likely also explains the preference of developers for frameworks built specifically for the high-volume, permissionless, low-latency demands of decentralized Web3 social media (e.g. [ActivityPub](#), [AT Protocol](#) or [Polygon](#) among others). [.xyz](#) (add the link of each in place like the rest)

5.3 Data Storage & Different Designs, or "Architectures of Different Existing Systems", or choose your own

Although hybrid adoption models offer partial solutions (e.g. using blockchain for identity, reputation and metadata identification, yet relying on decentralized non-blockchain peer-to-peer networks to store heavy data), they sacrifice some decentralization by transferring trust to storage owners. For example,

federated non-blockchain networks such as Bluesky (bsky.app) and, to a lesser degree, Mastodon (mastodon.social) accordingly remain more viable today for basic decentralized social networking needs, since they better handle massive data volumes than blockchain-driven ecosystems such as Farcaster (farcaster.xyz), Hive (hivesocial.app) or Lens Protocol (lens.xyz). Storing heavy media directly on a blockchain is particularly problematic, since these networks are designed for consensus on small, immutable data sets replicated across thousands of nodes; large files would basically bloat the chain, dramatically raise storage costs and slow validation. In this regard, standard content delivery networks or decentralized media hosts (e.g. IPFS, Arweave or other customized P2P layers) are therefore typically employed alongside blockchains (Arslan & Goker, 2022; Diba et al, 2025), with blockchain better suited to store metadata identifiers that indirectly link such content to its original media, eventually ensuring its immutability and verifiability. add a ref (or embed the actual link in text) with first use

“Cross-referenced ledgers can help ensure that users are acting intentionally or making credible contributions, not just casual or anonymous” (Interview 23)

This recorded transaction data, representing contextual information on user interactions, could be published automatically as open data on platforms run by non-governmental organizations for reuse by other stakeholders, for example, by independent developers or technically savvy citizens building consortium projects to measure public engagement across sectors. Blockchain-driven data management algorithms also offer more effective mechanisms to generate incentivized user feedback, particularly through transparent open source solutions (Petryk, Qiu & Pathak, 2023), aiming to generate public trust (Althero, Syahreza & Ortiz, 2023; Zhan, Xiong & Xing, 2023) in incentivized decentralized social media transactions and encouraging more active civic participation in public affairs.

A growing number of decentralized social networking algorithms focus specifically on enhancing incentivized peer-to-peer citizen participation, e.g. Sapien (sapien.network) or Mirror (mirror.xyz), aiming to facilitate more genuine interaction through economic incentives (Guidi & Michienzi, 2022; Delkhosh et al, 2023), partly reflecting the ideological motivation of technically savvy citizens to make transactions more independent of corporate actors through distributed, user-controlled interaction. More reliable and comprehensive blockchain-driven data management algorithms further offer new opportunities to analyze decentralized social media content, e.g. communication and market research, stakeholder and system analysis, benchmarking, comparative metadata analysis, generating public values such as transparency, immutability and accountability in information management practices.

6. Blockchain and data management techniques

Despite growing research on digital solutions intended to encourage more meaningful and genuine public interactions in social media environments, important questions remain concerning the mechanisms through which blockchain-based technologies may influence the authenticity, traceability and governance of civic participation, particularly how such systems might address AI-driven manipulation of digital expressions by humans (Kassen, 2026a). This matters increasingly since managing social networking activity depends on the interaction between human participation, automated systems

and platform governance; blockchain-driven approaches may offer new possibilities for managing and tracing social media data, yet their adoption also involves significant usability barriers, technical complexity and data governance challenges. This section therefore examines the potential of blockchain-based systems to enhance the verifiability and management of digital participation, given the socio-technical constraints that may limit their effectiveness.

“Data is crucial as it gives insights about behavior of the audience in social networks, performance of campaigners and helps to measure market trends. A really a big deal...”
(Interview 6)

These issues matter for understanding how decentralized, incentive-based mechanisms may influence the tracking and interpretation of public reactions in social media, including knowledge of blockchain protocols, data management practices, incentive and reputation mechanisms, and obscure algorithmic ecosystems that can directly influence content visibility and circulation.

“Average person may find decentralized platforms confusing, intimidating even. Many unfamiliar concepts; need to understand what platform tokens are, smart contracts, wallets...” (Interview 26)

“I believe there is a steep learning curve which can be a significant barrier...” (Interview 21)

This learning curve may be significant, though managing traditional centralized platforms can present even greater challenges, since their underlying infrastructures and algorithms are typically proprietary. This matters for information management because the intrinsic architecture of centralized platforms facilitate anonymity, duplication of digital actions and repeated low effort engagement, possibilities now amplified by automated content generation and algorithmic recommendation systems, devaluing civic engagement and undermining the capacity of civil society to inform and influence public affairs, a fundamental basis of representative democracy. Blockchain-based systems, with their decentralized data management mechanisms, offer alternative models to identify participants, verify their actions and commitments, and, more importantly, limit AI-driven manipulation, potentially making digital interactions among people more genuine, though effectiveness ultimately depends on the broader sociotechnical and data governance arrangements in which they are embedded in (Shin & Ibahrine, 2020).

7. Potential benefits of blockchain-driven social networking: reducing the cost of trust generation

The generation of public trust in public communications and transactions has always been a painstaking process (Shin & Bianco, 2020), requiring substantial regulatory, public relations and stakeholder engagement efforts and more so within the synthetically amplified environment recently created by AI-enhanced social media platforms, given the evasive, sometimes manipulative character of centralized interactions there. Centralized control and algorithmic content recommendation, governing the flow of

data and metadata and thereby enabling automatic censorship, retroactive redaction or outright bans of public figures over ideological disagreements by relying on repetitive patterns, undermines public trust and, to some degree, promotes slacktivism, which prior to the AI era carried a more positive connotation, manifesting instead in distorted public reactions to disinformation and manipulation (Saeidnia et al, 2025).

“Algorithms, centralized control may directly disrupt or interfere with the flow of information, could promote the spread of false or biased content. Blockchain makes it easy to verify content...” (Interview 21)

“It [blockchain algorithm] increases transparency, which in turn may help to generate trust in online settings, in social, networking interactions there.” (Interview 7)

Cure often becomes poison when applied in excessive quantities: public trust can be easily eroded when policymakers or opinion leaders communicate technological innovations without sufficient care or transparency, and generating it therefore remains a resource intensive process, particularly within an omnipresent digital environment given the sheer number of people engaged in social media daily. Blockchain offers a relatively low cost infrastructure to support trust in this area (Mlika, Karoui & Romdhane, 2024): by creating verifiable records of digital interactions and e-participation, it may strengthen the credibility and traceability of public engagement, including e-democracy and other digitally mediated transactions (Diba et al, 2025). Technical reliability alone, however, does not establish public trust, users must also trust the people and institutions that design, interpret and govern these systems.

“You cannot just trust algorithms, you have to trust people who develop and understand them... applying it, that’s the real challenge.” (Interview 19)

This observation highlights an important limitation of technology centered approaches to trust: blockchain may reduce certain verification and record-keeping costs, but trust in blockchain-based civic participation eventually depends on the transparency, competence and accountability of the actors developing and applying it. Its contribution to e-democracy therefore lies not in replacing human trust with algorithmic trust, but in combining verifiable technical mechanisms with credible human and institutional governance, warranting continued prototyping of civic engagement algorithms and protocols among academia, practitioners (Murray-Rust et al, 2023) and non-governmental organizations (Diallo et al, 2024). Generating public trust in decentralized, blockchain-driven social media accordingly involves addressing several consecutive challenges: transparency and accountability, security and data control, for instance, such platforms are preferable when built on open source code, thus auditable, inspectable and verifiable by any node in the network.

Such mundane yet critical technological features ensure the absence of intentionally or unintentionally hidden vulnerabilities to manipulate user data or content (Norbu et al, 2024). Peer-to-peer block creation, moreover, requires transparent, communicated rules governing how blocks are created, hash

information calculated and proliferated, and data-driven decisions made in a decentralized manner. In this regard, the maintenance of decentralized networks and single ledgers further provides ideal grounds for audit trails and cross-referenced metadata transactions, inspectable by any peer, or even onlookers, retroactively, where encrypted, decentralized management eliminates any single point of failure in data breaches or leaks.

8. Algorithmic blockchain-driven data management: peer-to-peer identification and use of cross-referenced metadata identifiers in social networks

Since blockchain ledgers will represent all citizens eligible to participate in important public activities, e.g. e-voting, e-petitioning, e-decision-making and other e-participation platforms, as well as legal entities (e.g. public organizations or even political parties), it is advisable to create a digital ecosystem of individually run ledgers promoting reputation-based content in a distributed manner via special metadata identifiers. One could imagine, for instance, an environment driven by blockchain ledgers where content is tied indefinitely to its creator, carrying an identifying tag that proliferates further through the network as a peer engages in decentralized social media interactions.

"The [blockchain] algorithm reduces centralization; no single entity controls the platform, so it promotes accountability, competition among content creators" (Interview 8)

Regarding the prototyping of such blockchain-driven social media networks, development could concentrate on a wide range of individual civic ledgers (CL), technically interconnected not only through peer-to-peer extension (e.g. blocks belonging to various peers, P1, P2, P3...PX) and preservation of recorded information (e.g. data recorded to public block P1.1 belonging to peer P1), but also through cross-referencing of metadata, with the use of separate individual ledgers ensuring more secure and reliable retention of e-participation data.

When we say "peer to peer" in blockchains, we usually mean by peer the decentralized nodes validating and maintaining the blocks; i.e., not the users generating the data. so, which of them you are referring to above and in the figure? I believe you are preferring to a hybrid permissioned design when peers are the data generators? P here is an entity gathering certain slacktivism activities in its blocks?

"Genuine interactions are manifested, distribution matters [...], yes, in a distributed way, which is possible through the use of individually run ledgers and metadata they generate" (Interview 10)

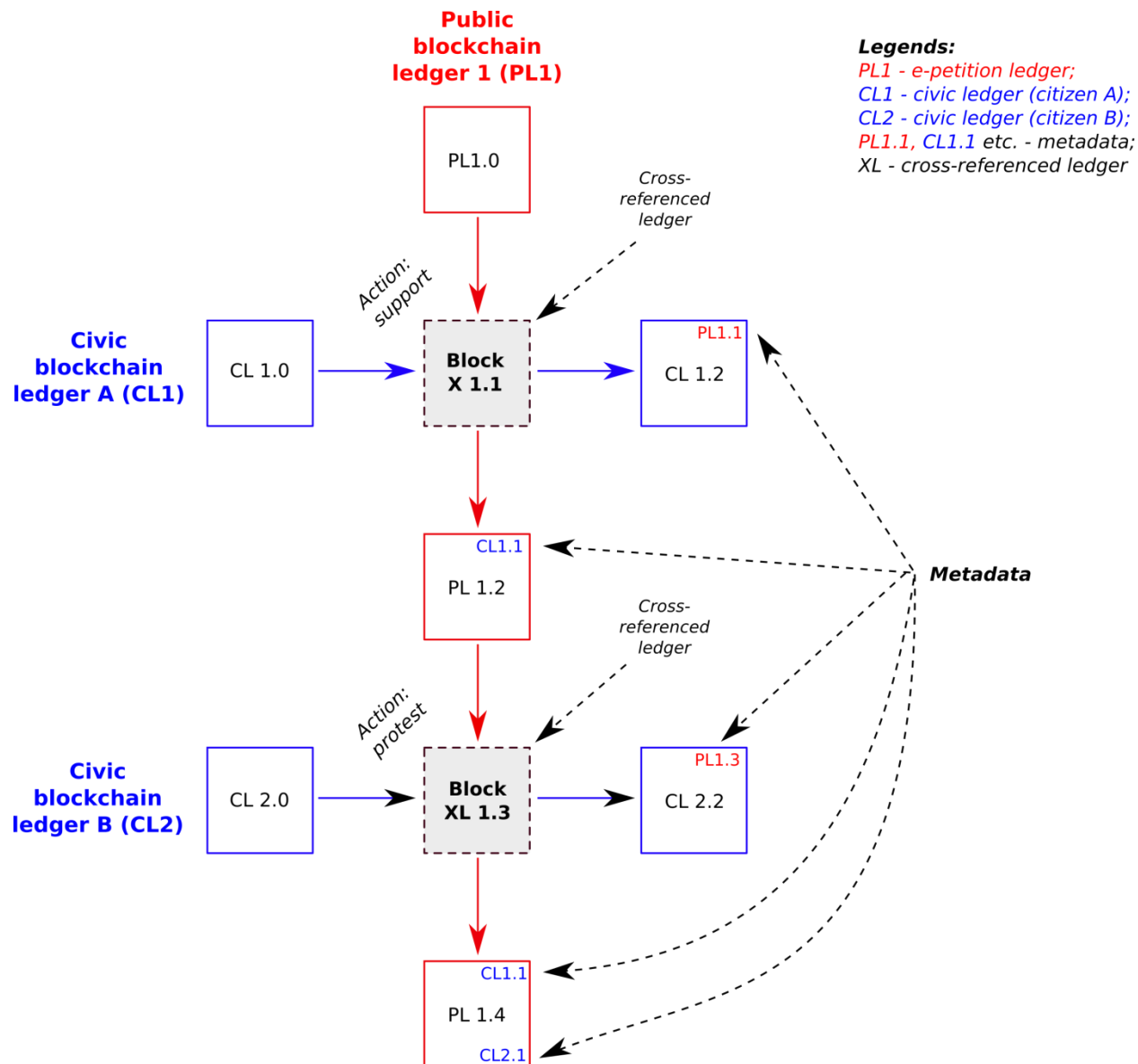
The figure is explaining it better, I cannot guide you on how to exactly phrase your explanation here. It seems to me that simple words on the proposed design that covers the following: 1-Each entity or activity is assumed to hold its own ledger (PL, p here for petition), 2-each citizen (maybe according to the platform he uses or to keep it local and save space) will have his own civic ledger (CL), 3- due to interoperability (CLs and PLs can each be of different type according to their vendors) we will need a cross-reference ledger (X or B) in the figure

Metadata of block owner P1, for example, directly affects hash information in block owner P3 belonging to a cognominal citizen or peer; since these personal civic ledgers could be used by citizens engaged in public participatory activism, e.g. public support, protest or awareness campaigns, cross-referencing and peer-to-peer communication generate a large volume of metadata across decentralized social media platforms (see figure 4).

In your figure 4 "block owner P1" is not a citizen. This fits figure 5 which seems like another design option. So, re-arrange and clarify

Figure 4. Linking citizen action to public ledgers: a metadata cross-referencing model

I wrote all of this before I reached sec.10, so merge the comments together.



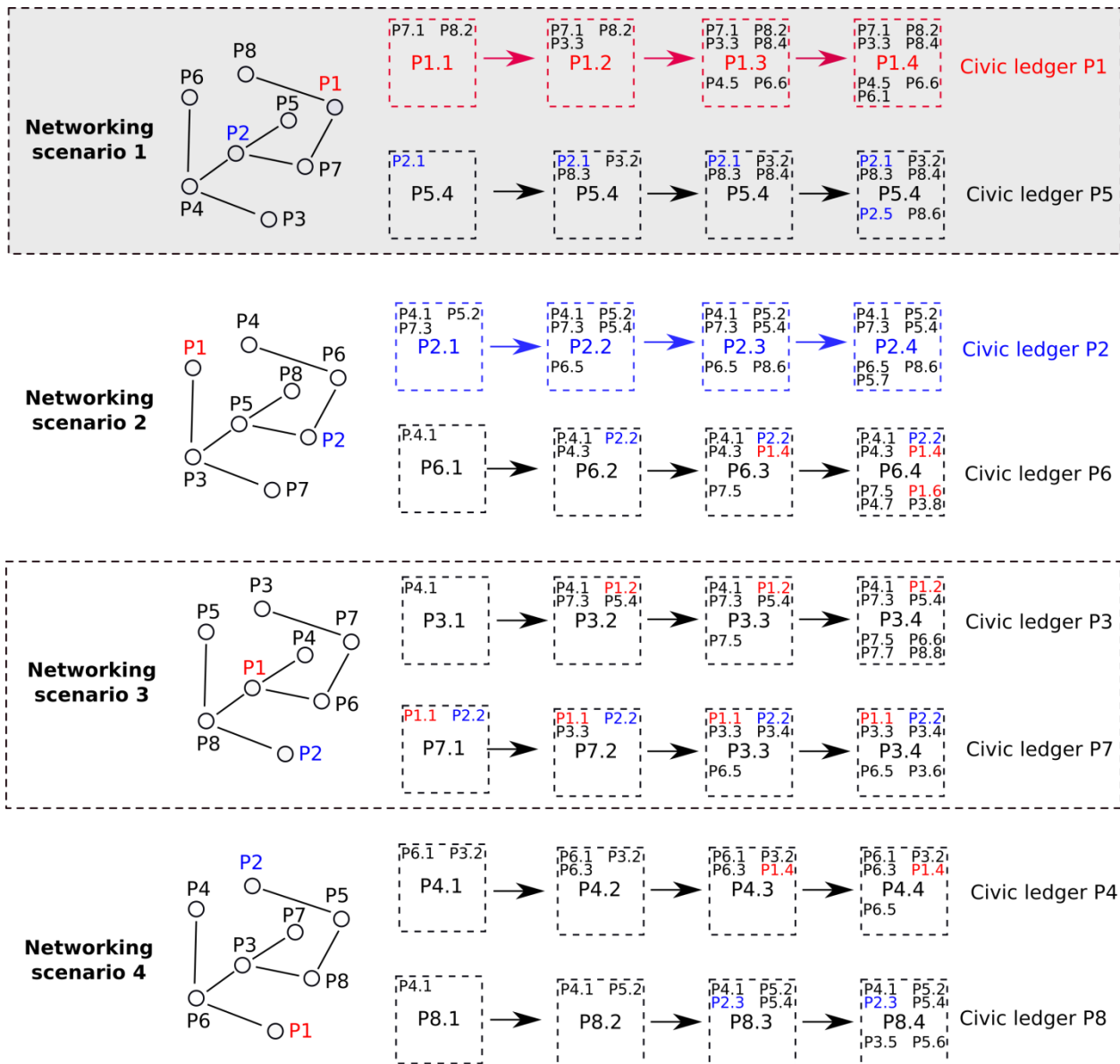
Source: own illustration based on the network analysis of key takeaways from focus group discussions

The proliferation of blockchain metadata could effectively curb the drawbacks of centralized data management, since double spending becomes impossible and retroactive redaction becomes exponentially difficult once cross-referenced metadata proliferates across permissionless or hybrid permissioned mechanisms, providing an additional layer of protection against manipulation; comments or activities tagged with a specially designed cryptogenic token thereby signal their importance to peers. the word "cryptogenic" could be mixed with "cryptographic" and not common in this context. in addition to assign different values (as coins/tokens) to different activities as mentioned earlier, certain significant activities could be assigned a distinguished special token.

"Once content is written, especially if it is cross-referenced across nodes or blocks, it is very hard to alter or delete it without detection. No tampering with data." (Interview 21)

One can visualize a hypothetical decentralized network in which peers (P1–P8) maintain individual civic ledgers storing metadata identifiers (e.g. P1.1, P2.3) that track the origin and proliferation of activities, e.g. posts, verifications, redactions; arrows in Figure 5, for example, illustrate how metadata propagates (e.g. P1.1, P1.2, P1.3), creating cross-referenced audit trails. Such a data-driven ecosystem promotes trust by enabling peers to verify content authenticity through consensus, identifying inconsistencies across ledgers (e.g. mismatched P1.4 in CL P4 versus CL P1) to flag potential fraud, where consistent chains validate genuine interactions. Metadata identifiers thus function as digital breadcrumbs, commonly used in blockchain-driven forensics (Trozze, Davies & Kleinberg, 2023), enabling decentralized networks to distinguish genuine from fraudulent activity by tracing its proliferation across multiple independent ledgers. demonstrate the inconsistency error you mean on the figure, not clear

Figure 5. Peer-to-peer civic ledger structures and cross-referenced metadata propagation



Source: own illustration based on the network analysis of key takeaways from focus group discussions

9. Projecting key risks and challenges of incentivized blockchain-driven social networking

9.1 Regulatory oversight: responsible AI, blockchain and accountability gap in decentralized networks

Traditional social networking platforms such as Facebook or X (formerly Twitter) act as centralized intermediaries between users, whether public, corporate or civic; governments accordingly engage directly with these entities to enforce laws, issue takedown orders or demand compliance with content moderation standards within a jurisdiction. Decentralized social networks, built on blockchain or peer-to-peer protocols, dilute this single point of control (Tatar, Gokce & Nussbaum, 2020): there is no

corporate headquarters, no CEO to summon, no central server to inspect, responsibility diffused across potentially millions of independent nodes globally, none individually controlling the network, leaving regulators facing a fundamental dilemma over whom to hold accountable when illegal content, e.g. hate speech, cyberbullying or defamation, appears, since without a clear legally responsible entity, enforcement mechanisms such as fines or sanctions become difficult to apply.

“Regulations usually hold platforms accountable for hosting content. If they fail to remove it upon notice, they are punished. Now let’s question this new model. Who exactly represent a platform?” (Interview 19)

Decentralized networks, such as Sapien Nation, Hive, Lens and many others, operate across national borders, with users, infrastructure operators, moderators and other participants potentially subject to different legal systems and cultural norms concerning acceptable speech. As a result, content classified as hate speech in one jurisdiction may constitute protected expression in another, creating significant jurisdictional difficulties for regulators, particularly where network infrastructure or participants are located outside national territory or operate under pseudonymous identities (Howard, 2019). The decentralized organization of these networks may further complicate enforcement because users can potentially access services through VPNs, alternative gateways or replicated interfaces, where distributed infrastructure can make it difficult for a single jurisdiction to impose comprehensive restrictions on network activity. Moreover, attempts to restrict or disable a decentralized network in one jurisdiction may encourage the emergence of alternative instances, forks or replicated services, thereby limiting the effectiveness of territorially bounded regulatory interventions and making consistent cross-jurisdictional enforcement considerably more difficult.

“There are actually a number of vulnerabilities... blockchain is not bulletproof, contrary to a common belief. It is not that robust technologically, not even conceptually.” (Interview 15)

Without centralized moderators, decentralized networks typically resort to community governance models, often token-based voting within DAOs, a process that can be inherently slow, e.g. reaching consensus on controversial content removal may take days, weeks or never occur, allowing flagged content to spread virally during deliberation; token-based e-voting mechanisms also create vulnerability to manipulation (Diba et al, 2025), since individuals or coordinated groups with disproportionate token holdings can sway decisions to protect harmful content or silence legitimate criticism. Though tokenized quadratic e-voting offers a partial solution (Kassen, 2025), the process may still lack the speed and trained judgment of professional moderation teams, leaving suspicious content active longer than on centralized platforms.

“These algorithms require substantial operational resources to fulfill [...], hardware, networks, infrastructure, knowledge, skills... A lot of variables.” (Interview 6)

Even where a tokenized vote decides to remove content, the technical reality often renders this superficial: content recorded on universal decentralized storage protocols such as IPFS or Arweave typically leaves traces across the network unless recorded via encrypted, multi-layer verification protocols, and other gateways can still access it via its unique content identifier, meaning taking down a reference on one application does not delete the underlying replicated data. Censorship attempts, moreover, can trigger the Streisand effect (Mach, 2022), as users deliberately replicate content in protest or fork the decentralized protocol entirely, amplifying legally non-compliant material.

add a ref (or embed the actual link in text) with first use

“Blockchain could help to curb it [polarizing content]. But I believe it also makes social media less attractive, less popular. That’s a problem.” (Interview 22)

Decentralized social networks, if they become more popular, will reshape the social media landscape entirely, dissolving traditional understandings of accountability. Namely, the core features enabling user empowerment and censorship resistance, i.e. lack of central control, immutability, global reach and quasi pseudonymity, create real challenges for applying legal frameworks designed for centralized platforms. Addressing harmful content in this new paradigm demands novel legal definitions of responsibility, privacy enhancing enforcement techniques, international regulatory cooperation and data governance models embedded within protocols themselves, i.e. solutions facing significant technical, political and philosophical hurdles that leave regulatory oversight considerably complicated.

9.2 Systemic barriers: public inertia and greater demand for increased public awareness

Engaging in tokenized e-participation requires citizens to learn how to spend such blockchain-driven social currencies wisely; within a market environment increasingly driven by AI (Huang, 2024) and automated blockchain solutions, emotions and reactions become the main denominator generating a new digital media currency based on the attention economy (Nicoli, Louca & Iosifidis, 2022; Zhan, Xiong & Xing, 2023), offering an interesting platform to experiment with universal income schemes built on this incentivized model, since automation will likely replace nearly everything humans do, leaving intact only what humans alone can do: expressing genuine emotion and reaction.

Using such a limited digital resource as social media tokens, which represent public trust as a quantifiable legal tender within decentralized platforms, is a critical step toward successful adoption of such technologies. Public awareness is especially crucial for services such as e-participation, e-petitioning, e-voting, e-collaboration, e-funding and e-decision-making, which become possible once citizens hold a limited number of non-fungible tokens to spend on significant public causes, transacted through participatory networking and public action, though this could also attract various interest groups to these networks.

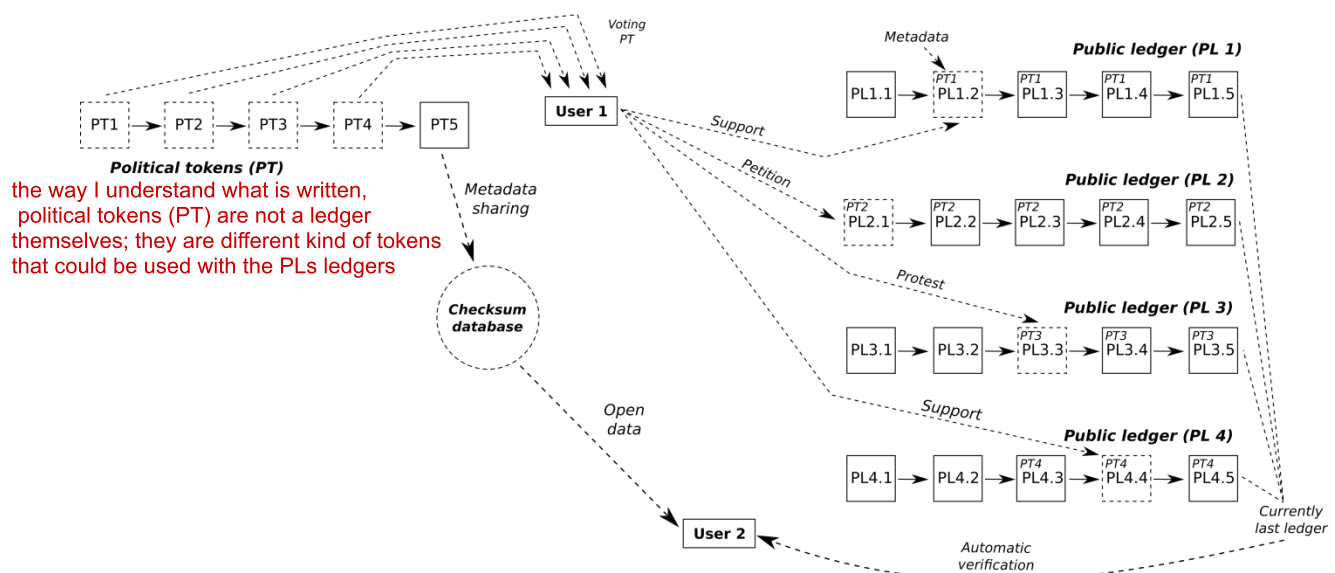
"non-fungible tokens (NFTs) are not exactly spendable; they're bought or sold. regular (fungible) tokens are the ones which are like money (platform coins) and can be spent on it or sometimes exchanged outside of it into regular money

Algorithms, furthermore, prioritize engagement since humans are naturally wired to respond to emotional stimuli, particularly outrage or conflict, a phenomenon amplified by negativity bias, especially evident online (Gazit et al, 2025); centralized platforms typically exploit this through machine learning that identifies and promotes emotionally triggering, attention-driven content (Robertson et al, 2023). In

this regard, the promise of blockchain transparency and tamper proof records might technically reduce AI-driven emotionally charged interactions and verify sources, but it does not address the core issue of social networking: the human craving for emotional peer-to-peer engagement. As such, in sanitizing platforms of neo-slacktivist interactions, i.e. removing amplified content and the chaos of anonymity, blockchain may in fact strip away the very public arena users find compelling to join or confront, even where this proves manipulatory or meaningless for the cause of public debate itself.

One could imagine, for example, a model to adopt differentiated blockchain-driven protocols to automatically verify public activities, e.g. protests, petitions or expressions of support, through a parallel ecosystem of dynamically generated political tokens (PT1–PT4) cross-referenced with public ledgers (PL1–PL4), whereby citizens spend PT tokens to record actions on specialized ledgers corresponding to different activity types (e.g. PL1 for support, PL2 for petition or protest), transactions cryptographically secured and chain-linked for tamper proof verification, with transparency further enhanced by publishing ledger checksums to an open repository for independent validation without exposing raw data. In this context, by combining tokenized e-participation, distributed ledger recording and quadratic verification, i.e. multi-layered validation across tokens and checksums, the model may thus promise immutability: a tamper proof mechanism resisting the data manipulation that actually fuels neo-slacktivism in the first place (see figure 6).

Figure 6. A model for automatic quadratic blockchain-driven verification of public activity via social media networking tokens and related generation of single ledgers



Source: own illustration based on the network analysis of key takeaways from focus group discussions

10. Discussion. Connecting conceptual models to practical applications

One of the reasons for the recent advent of blockchain-driven social networking is that traditional centralized platforms, amplified by AI-driven content algorithms, have facilitated algorithmic neo-slacktivism. Blockchain promises to counter this through cryptographic tokens and cross-referenced ledgers immutably tying digital interactions of real humans to verified identities: signing an e-petition, for instance, could generate an on-chain credential, unlocking higher-impact activities such as e-voting on community budgets or collaborative decision making, thus turning engagement from performative act into a verifiable audit trail. Fragmentation across technological standards, protocols and platforms, coupled with regulatory uncertainty, may nevertheless constrain the wider adoption of these systems by limiting cross-platform compatibility, increasing institutional and technical barriers, and reducing their capacity to respond effectively to evolving market needs.

10.1 Blockchain cross-referenced ledgers in social media and e-voting

Cross-referencing metadata across decentralized ledgers enables traceable participation ecosystems, as illustrated by the Decidim project (Spain), where civic proposals are cryptographically hashed onto a public ledger (The Decidim Project, 2025). Although Decidim, a citizen e-participation platform implemented in Barcelona, is not itself built on blockchain, it integrates blockchain-based voting solutions from Vocdoni, one of the most universally verifiable e-voting protocols (Haarseim, 2023). If further integrated with blockchain, other interactions, e.g. debates, comments and likes, could likewise be tokenized and linked to verifiable participation records; event attendance, for instance, could be confirmed through cryptographic proof from verified check-ins such as geolocation attestations and recorded via an Ethereum-based proof of attendance protocol or comparable credential. The Follow My Vote project (FMVP, 2025) offers a further illustrative example, aiming to strengthen public trust in digital credentials of participants.

In a social networking environment, empirical analysis of such decentralized platform as Steemit sheds light further nuance to this aspect of public trust generation. For example, curation earnings there have been shown to concentrate among a small cohort of high stake accounts rather than content creators (Ba, Zignani & Gaito, 2022), reminiscent, within a blockchain-native setting, with the amplification inequities associated with centralized platforms. Although the incentive structure of blockchain-based social media can shape user-generated content by linking content creation and curation to cryptocurrency rewards (Li, Loahavilai & Naktnasukanjn, 2022), bot-driven voting has been found to inflate apparent participation evenⁱⁿ blockchain-driven ecosystems (Guidi & Michienzi, 2020): illustrating precisely the synthetic engagement that cross-referenced civic ledgers and metadata identifiers are designed here to detect. Also, because rewards are denominated in a volatile native token, posting and curation behavior has proven sensitive to price fluctuations (Ba, Zignani & Gaito, 2022), underscoring how tokenized incentives inevitably interact with market forces that designers must anticipate to avoid its negative effects on these e-voting processes.

10.2 Tiered consensus schemes

As experts recommend, tiered and hybrid blockchain architectures can combine efficiency and security by selecting consensus mechanisms suited to different transaction categories: routine interactions (e.g. comments, likes, reposts) may rely on energy-efficient PoS consensus, as demonstrated by platforms such as Steemit and Hive, whereas more sensitive activities, including e-voting, fund disbursement and collaborative decision making, generally require stronger security and data governance guarantees through more robust or permissioned architectures. This is an important aspect of an anti-slacktivist strategy even in blockchain-driven ecosystems. The experimental blockchain voting initiative in West Virginia, for instance, employed cryptographic hashing and secure transaction verification to enhance ballot integrity without relying specifically on PoW (Miller, ²⁰¹⁸2025). It is interesting to note that the blockchain pilot in the Swiss city of Zug (ZBPP, 2018) illustrates how proof-of-concept projects can evaluate such feasibility. Although PoW continues to offer strong security in some public networks, its high energy consumption remains a limitation, one renewable-powered facilities such as Icelandic geothermal data centers may partially offset (Pan & Goldman, 2023). The Democracy.Earth project (DEEP, 2025), operating on Ethereum, offers a further example of differentiated architecture, integrating PoH as an application layer identity mechanism rather than a consensus protocol: users submit evidence, including a short video and community verification, to demonstrate unique human participation, reducing Sybil attacks and thus strengthening the credibility of decentralized participation.

this happened
2018, that's why
Ethereum was
POW at that time

10.3 Token economics for more inclusive e-governance

Token-based incentive systems encourage more meaningful, sustained civic engagement within decentralized governance ecosystems. In online philanthropic activities, quadratic funding, for example, allocates matching grants according to community preferences, giving proportionally greater influence to numerous small contributions than to a few large donations, encouraging broad-based engagement of users and reducing the influence of concentrated financial resources: blockchain enhances this mechanism by providing a transparent, immutable record of donations, allocations and voting outcomes. Token economies can also strengthen civic engagement through reputation-based incentives: soulbound tokens (SBTs), proposed by Ohlhaber, Weyl and Buterin (2022), are non-transferable digital credentials representing achievements or civic contributions in contrast to financial assets blockchain users usually accrue, reducing speculative behavior and allowing individuals to accumulate verifiable records of civic engagement, potentially forming part of a lifelong digital civic identity. Such tokenized data governance nonetheless introduces certain challenges: financial incentives may unintentionally encourage superficial participation or token farming, where concentrated token balances raise concerns about political inequality and even digital vote buying. Mitigating these risks requires combining financial incentives with non-transferable reputation credentials, transparent governance rules and identity verification mechanisms such as the earlier discussed PoH.

DAO-based content moderation represents a further important application of tokenization in the area: platforms including Minds, Hive, Peepeth and Sapien incorporate token-based mechanisms rewarding users for contributing to governance through content creation, moderation and identification of misleading information, strengthening community-driven moderation, discouraging low quality reporting, for instance, rotating fact checking DAOs could audit disputed claims, rewarding accurate

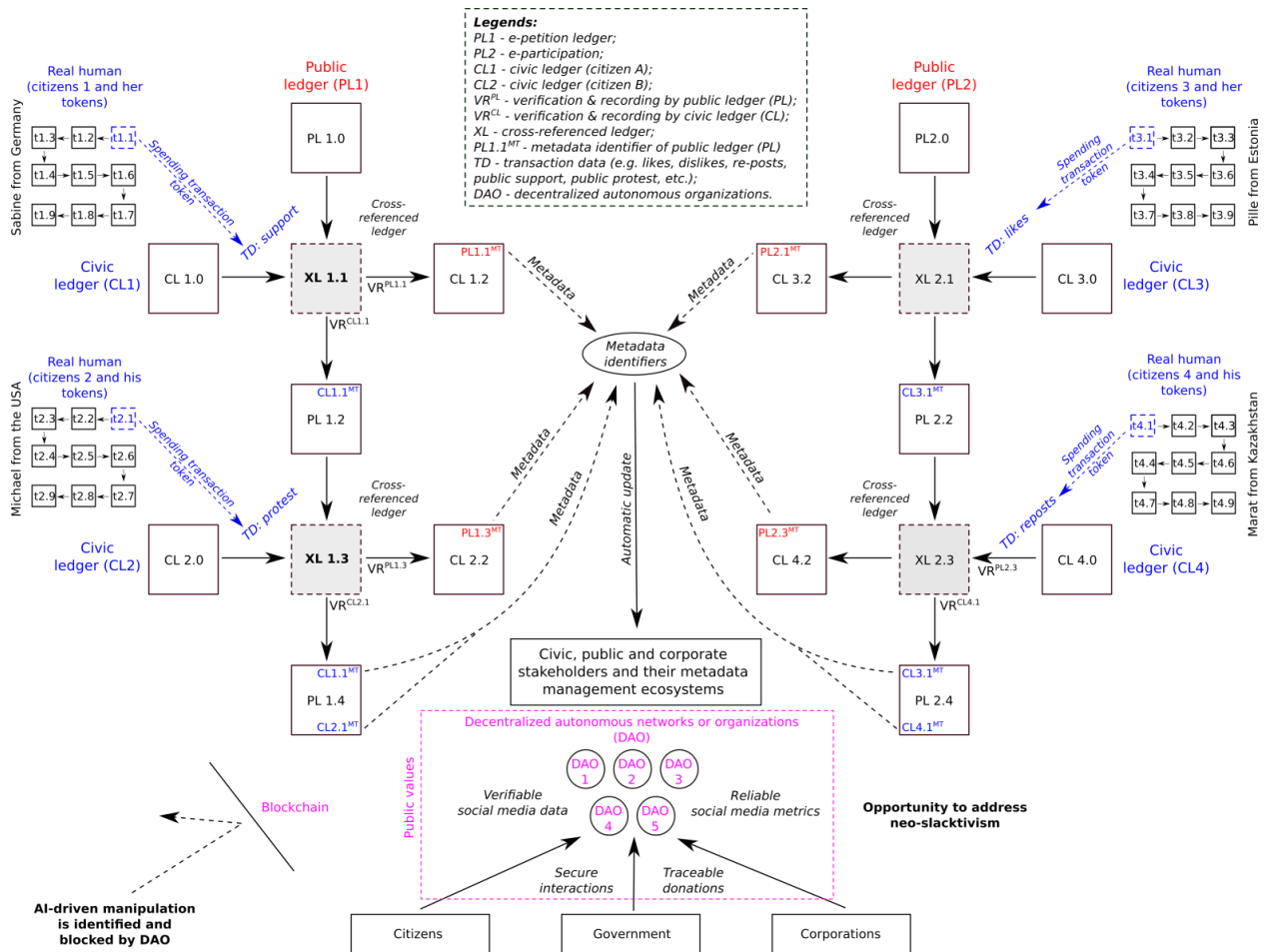
assessments with tokenized credit, though such mechanisms must carefully promote engagement, navigating potential strategic behavior and token farming by users. Complementing this, blockchain-supported content provenance can further enhance information integrity: The Project Origin (2025) demonstrates how cryptographically verifiable metadata can establish provenance of digital media by linking content to its original creator, using a C2PA protocol (Coalition for Content Provenance and Authenticity) that certifies the source and genuineness of public information. Thus, combined with blockchain ledgers, such provenance information can improve transparency, facilitate detection of AI-generated content and strengthen public trust in decentralized ecosystems.

10.4 Decentralized moderation of disinformation and models of technology adoption

Illustratively, blockchain-driven ecosystems may combat AI-driven neo-slacktivism by creating verifiable proof of genuine human engagement, via a hypothetical dual layer ledger system in which personal civic ledgers (CL1–CL4) serve as cryptographically signed identity vaults, each containing tamper proof transaction data (TD) for individual interactions (e.g. TD: protest, TD: support) linked to unique metadata identifiers (CLx.y^{MT}) and interlocking with public ledgers (PL1/PL2) governing e-petitions (PL1) or e-participation (PL2) through cross-referenced ledgers (block XL). When a citizen initiates an action, e.g. signing a petition or participating in e-voting, the system executes: a) a CL verification code (VR^{CL}), validating the action via zero-knowledge proofs to confirm human agency without exposing private data; b) a PL verification code (VR^{PL}), generating a public record with a metadata watermark (PLx.y^{MT}); and c) a XL bridging code, creating an immutable cross chain proof (e.g. block XL1.1) binding the CL identity token to the PL public action, i.e. a triple-layer verification isolating synthetic engagement by requiring consensus across all three ledgers. this explanation perfectly describes figure 4 and should be written before it

This architecture may directly combat AI-driven manipulation by transforming superficial interactions into resource intensive transactions, namely each like, repost or support action consumes non-transferable CL tokens of a citizen, forcing cryptographic commitments that expose bot patterns through *token exhaustion*, i.e. progressive depletion of an actor's (real human or bot) token balance through repeated costly actions, rendering sustained high-frequency automated behavior economically unsustainable, thus consistent with resource-based Sybil resistance approaches in blockchain systems (Dong, Feng & Wang, 2026; Lai et al, 2026); suspicious TD patterns (e.g. 10000 likes within seconds) are flagged by DAO-governed algorithms, and discrepancies between CL private intent and PL public action trigger automated sanctions. In this regard, by tying digital activism to tokenized identity, the system thus replaces low-effort neo-slacktivism with provable human interaction: citizens build verifiable reputation, whereas corporations, governments and NGOs access auditable engagement metrics, rendering AI-generated manipulation of data computationally prohibitive and easily detectable (see figure 7).

Figure 7. A concept of blockchain-driven social interactions via cross-referenced ledgers and metadata identifiers



Source: own illustration based on the network analysis of key takeaways from focus group discussions

turn this paragraph into a detailed subsection, failure cases should be backed by references explaining the failure reasons the elaborate on how could be or was avoided in later ones to not eliminate the whole idea of using blockchains

Moderation speed, however, could remain slow due to technical limitations of blockchain networks, suggesting hybrid AI-human ecosystems that prioritize AI detection, reserving final decisions on suspicious content for token-weighted human votes. Blockchain-driven algorithms proposed toward this end, such as the Dirt (2018) and Pressland (2019) protocols, aimed to increase transparency and trust in digital media and journalism, though both ceased to exist in the early 2020s amid unsustainable funding and the broader cryptocurrency crisis of that period. As a result, paradoxically, corporate centralized social media settings, as such, still remain superior in community-driven flagging of disinformation, as

exemplified by Community Notes on X (2025). Also, community notes have papers studying them and should be mentioned for comparison, since X does not use blockchains.

you could also add Facebook as an example, from a famous Cyber Security paper "Bugs in Our Pockets":

"One incentive for firms to adopt end-to-end encryption may be the costs of content moderation. Facebook alone has 15 000 human moderators, and critics have suggested that their number should double [15]."

11. Conclusion

[15] is Barrett P. Who moderates the social media giants?. 2020. <https://www.stern.nyu.edu/experience-stern/faculty-research/who-moderates-social-media-giants-call-end-outsourcing>, check also [45] <https://about.fb.com/news/2019/09/combating-hate-and-extremism>, and have a look at [26]

As an agenda setting article, this piece aims to initiate further debate on the potential of blockchain-driven algorithms to better manage user activity data in social networking interactions, particularly regarding solutions capable of addressing the growing advance of AI-driven information manipulation as a technological, political and socioeconomic phenomenon of great concern. As discussed throughout, decentralized blockchain-driven data management represents one of the most promising areas in which

References analyzing Community on X:

(ACM 2024) <https://dl.acm.org/markets/2019/01/11/0-success-why-blockchain-apps-just-arent-taking-off>, (CSOH 2025) <https://www.csohate.org/2025/06/30/x-community-notes-south-asia/>, (UW Nov 2025) <https://www.washington.edu/news/2025/09/18/community-notes-x-false-information-viral/>

Failures:

Dirt (<https://share.google/aimode/UTaujbh7nD4GZgFt>), PressLand & other failures (<https://share.google/aimode/Ln4egla6K12AR4R9a>) check embedded links in each <https://www.coindesk.com/markets/2019/01/11/0-success-why-blockchain-apps-just-arent-taking-off>

<https://www.hbs.edu/faculty/Pages/item.aspx?num=57393>

if you tried the Dirt link protocol you have cited, you will find that it is now acquired by "Guestinvest"

distributed ledger technology render these interactions more genuine, ensuring traceability of decentralized social media activity and greater personal responsibility among peers or block owners through specially designed cryptographic tokens, cross-referenced metadata and related content identifiers. These insider perspectives on technology adoption may help key stakeholders better understand the public value of digitally enhanced communication, raising academic and public awareness of AI-driven information sharing ecosystems and enabling meaningful, sustainable participation in important public causes they facilitate.

In this regard, key findings reveal that blockchain offers promising mechanisms to enhance genuine digital activism through cryptographic tokens, permissionless ledgers and cross-referenced metadata identifiers, which collectively tie content to creators, validate unique contributions and curb AI-driven manipulation (e.g. disinformation, synthetic engagement, data manipulation). Significant challenges nonetheless impede adoption: systemic risks include fragmented regulatory frameworks, low public awareness of tokenized participation and vulnerability to social engineering, whereas technological barriers involve high computational demands (particularly for PoW consensus) and the need to balance decentralization against efficiency (e.g. favoring PoS or DPoS for routine actions such as comments or likes yet reserving robust PoW for critical activities such as e-voting). Experts accordingly emphasize that success requires more open source protocols, differentiated consensus mechanisms and further algorithmic testing, through which blockchain create accountability layers: verifying human involvement, tracking real-world impact and ensuring resources flow to meaningful interaction.

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